

GOVERNMENT OF JHARKHAND • STATE INNOVATION DESK

OFFICIAL WORKFLOW SPECIFICATION

JHARKHAND SOCIETAL INNOVATION & COLLABORATION PLATFORM (JSICP)

CLOSED-LOOP GOVERNANCE

9-Step End-to-End Operational Workflow: From Grassroots PRIs to Apex HEIs & CSR Escrow

NEP 2020 & CSR S.135

Executive Framework: JSICP operationalizes a 3-sided closed-loop innovation pipeline uniting **Grassroots Citizens & PRIs**, **4 Apex Higher Education Institutions** (BIT Mesra, IIT ISM Dhanbad, AIIMS Deoghar, BAU Ranchi), and **Industry/CSR Anchors** (Tata Steel CSR Foundation) under the authoritative governance of **State Apex Command**. Below are the **9 foundational operational steps** governing every societal challenge from submission to verified village deployment.

STEP 01	Grassroots Problem Reporting & PRI Verification Actors: Citizen / Jal-Mitra / Gram Panchayat Mukhia Citizens, Jal-Mitras, or Gram Panchayat Mukhias submit unaddressed societal challenges in local languages (Hindi, Nagpuri, Santali, English) via mobile web or WhatsApp bot. Submissions capture voice dictation, high-resolution geotagged photos, and village PRI verification. Portal Endpoint: /portal/citizen & /submit Primary Output: Official Ticket (e.g. JSICP-2026-2481) Data Captured: GPS Geotag + Voice Audio + Visual Evidence Audit Stamp: Panchayat Ingestion Verified
▼ Automated Ingestion into State AI Problem Engine ▼	
STEP 02	AI Problem Engine: Priority Scoring & 5km Deduplication Actors: State AI Engine (Vision ViT + NLP Processor) Multi-modal automated analysis computes a composite Priority Score (0–100) based on severity weight (38%), affected population density (24.5%), district vulnerability index (18%), and SDG alignment (14%). Performs spatial Haversine 5km vector deduplication to merge cluster complaints. Core Algorithms: Haversine 5km Radius + ViT CV Tags Scoring Metric: Weighted Priority Index (0 to 100) Deduplication: Zero Duplicates in 5km Radius Verified Status Code: under_ai_review → pending_nodal_review
▼ Priority-Scored Ticket Evaluated by Explainable AI Matrix ▼	
STEP 03	Explainable AI (XAI) Academic Routing Matrix Actors: Explainable AI Decision Intelligence Gate Domain NLP and institutional capability indexing evaluate specialized laboratory infrastructure across Jharkhand's 4 Apex HEIs: • AIIMS Deoghar (98% Fit): Viral Fever, Epidemic Response, MedTech • IIT (ISM) Dhanbad (99% Fit): Coal Fire, Methane Vents, Geo-Robotics • BIT Mesra (96% Fit): Fluoride/Arsenic Water, Environmental IoT • BAU Ranchi (98% Fit): Lac Rot, Farm Automation, Agronomy XAI Framework: Explainable AI Decision Intelligence Gate Ranking Depth: Top-3 Matched Higher Education Desks Matching Score: Domain Fit Percentage (96% – 99%) Audit Trail: NLP Key Terms & Research Lab Justification

EXECUTIVE OPERATIONAL WORKFLOW		
PHASE 2: STATE APPROVAL, INSTITUTIONAL INTAKE & BILATERAL MOU		
STEP 04	State Apex Command & Nodal Approval Gate State Administrative Command inspects the priority breakdown, NLP keyword reasoning, and computer vision flags within the <i>Human AI Validation Gate</i>. With single-click executive authority, the IAS officer approves and officially routes the challenge to the designated apex university desk.	Actor: State Nodal Officer (Dr. Shailesh Kumar, IAS) Portal Endpoint: /portal/admin (Validation Gate Tab) Authority: State Nodal Officer / IAS Director Action: Inspects XAI Rationale & Confirms Routing Status Shift: pending_nodal_review → routed
▼ Officially Routed to Institutional University Nodal Desk ▼		
STEP 05	University Nodal Desk Acceptance & Faculty Allocation The University Nodal Officer reviews the routed challenge on the institutional dashboard (/portal/hei-nodal), verifies departmental research fit and laboratory capacity, officially accepts the ticket, and assigns a specialized departmental Professor as the Lead Faculty Research Mentor.	Actor: University Nodal Officer (AIIMS / IIT ISM / BIT / BAU Desk) Portal Endpoint: /portal/hei-nodal Institutional Scope: Ecosystem Isolation (AIIMS / IIT / BIT / BAU) Action Taken: Accept Challenge & Assign Faculty Mentor Status Shift: routed → accepted_by_hei
▼ Assigned to Departmental Faculty Mentor Workspace ▼		
STEP 06	Faculty Mentor Scoping & Student Cohort Assembly The Faculty Mentor accesses /portal/faculty to structure the engineering response: 1. Team Assembly: Pairs multidisciplinary student innovators from their institution (Hardware Lead, IoT/Software Lead, Field Liaison). 2. R&D; Proposal: Formulates technical methodology, 5-stage milestones, bill of materials, and budget grant requirement.	Actor: Lead Faculty Research Mentor & Lab Director Portal Endpoint: /portal/faculty Student Cohort: Multidisciplinary Institutional Student Team Deliverables: Technical Proposal + 5-Stage Milestone Roadmap Funding Mode: Declares if CSR Grant Co-Funding is Required
▼ Proposal Published for Industry / State Funding Clearance ▼		
STEP 07	Funding Route: Industry MoU or Direct State Track Path A (Industry / CSR Grant): Listed on /portal/industry. Tata Steel CSR evaluates technical feasibility, executes a legally binding Bilateral MoU with digital E-Sign, and commits matching tranche grant into Escrow. Path B (Direct State Track): Fast-tracked via Academic Seed Corpus without external corporate MoU.	Actor: Industry Partner (Tata Steel CSR) / State Innovation Corpus Portal Endpoint: /portal/industry (Marketplace) Legal Instrument: Bilateral Tripartite MoU Digital E-Signature Financial Lock: Smart Contract Escrow Grant Commitment Status Shift: proposal_drafted → mou_executed / funded

PHASE 3: LAB FABRICATION, DUAL SIGN-OFF & CITIZEN VERIFICATION

CLOSED-LOOP COMPLETION

STEP 08

Student Lab Prototyping & Document Vault Upload

Student team enters the active Sprint Workspace (/portal/student). Team leads delegate sub-tasks to member specialists. The cohort fabricates physical prototypes in university laboratories (PCB etching, CAD modeling, sensor firmware, chemical testing) and uploads telemetry logs and PDF test sheets into the Document Vault.

Actor: Multidisciplinary Student Innovator Cohort

Portal Endpoint: /portal/student (Sprint Kanban & Vault)

Role Division: Team Lead Delegation vs Member Task Execution

Vault Artifacts: PDF Laboratory Test Sheets + Telemetry Streams

Verification: Progress Slider % & Milestone Completion Claims

▼ Milestone Deliverables Pushed to Two-Tier Approval Gate ▼

STEP 09

Milestone Dual Sign-Off Gate, Blockchain Escrow & Citizen Verification

1. Gate 1 (Faculty Academic Stamp): Professor inspects uploaded test sheets on /portal/faculty and applies official Academic E-Sign.

2. Gate 2 (District Officer / DM Stamp): Government District Officer audits ground telemetry on /portal/admin and applies District Stamp.

3. Escrow Release: Blockchain Smart Contract triggers automated tranche payment to university research account.

4. Citizen Loop Closure: Prototype installed on-ground in village. Citizen's 10-step stepper marks COMPLETED, unlocking the 5-Star Ground Impact Survey.

Actor: Faculty Mentor • District Officer (DM) • Citizen / PRI

Dual Sign-Off Desk: Faculty Academic E-Sign + District DM Stamp

Financial Release: Polygon SHA-256 Escrow Tranche Disbursal

Field Deployment: Physical Commissioning at Gram Panchayat

Final Status: deployed → verified_closed (5-Star Citizen Rating)

END-TO-END GOVERNANCE & RBAC ENDPOINT AUDIT MATRIX

Lifecycle Phase	Primary Stakeholders	Portal Endpoint	Key Governance & Verification Gates
Steps 01 – 03 Triage & AI Routing	Citizen, Jal-Mitra, Gram Mukhia, State AI Engine	/portal/citizen /submit	5km Geospatial Deduplication + ViT Priority Score (0–100) + Explainable AI (XAI) Institutional Fit Ranking
Steps 04 – 05 Nodal Institutional Gate	State Secretary (IAS) & University Nodal Officers	/portal/admin /portal/hei-nodal	Human AI Validation Gate Approval + Institutional Acceptance + Departmental Faculty Research Mentor Assignment
Steps 06 – 07 Scoping & CSR Funding	Faculty Mentor, Student Cohort, Tata Steel CSR	/portal/faculty /portal/industry	Multidisciplinary Team Builder + 5-Stage Proposal + Bilateral MoU Execution with Digital E-Signature & Escrow Lock
Steps 08 – 09 Dual Sign-Off & Closure	Student Team, Faculty, District DM, Citizen	/portal/student /track	Gate 1 (Faculty Stamp) + Gate 2 (DM Stamp) → Blockchain Escrow Tranche Disbursal → 5-Star Ground Impact Survey

OFFICIAL VALIDATION STAMP: Verified by Government of Jharkhand Higher & Technical Education Command. All transactions cryptographically sealed on SHA-256 Distributed Ledger. Valid for State Innovation Submissions & SIH 2026.